

Electronics Task:

Task1:

1.1:

Load A (RE22 Relay) → 9 V, 40 mA

Load B (HC-SR04 Ultrasonic Sensor) → 5 V, 15 mA

1.2:

A step-down regulator is needed because the 12.6 V battery voltage is higher than the voltage required by both loads (9 V and 5 V).

1.3:

Yes, reverse-polarity protection is needed to protect sensitive circuit components and regulators from permanent damage if the battery is accidentally connected backwards.

2.1:

I will use an LDO for Load A because the voltage difference is relatively small and the relay has a low current requirement. For Load B, I will use a buck converter because the larger voltage drop from 12.6 V to 5 V would cause more power loss in a linear regulator.

2.2:

I'll pick P-channel MOSFET reverse-polarity protection circuit because it protects the circuit components and regulators if the battery is connected backwards, and it is preferred over a standard diode because it provides effective protection with very low V drop therefore reduces power loss.

3:

LDO for 9 V

Selected component: LM7809

Input voltage range: 11.5 V – 35 V

Maximum output current: 1.5 A

The selected LDO is suitable because its input voltage range covers the battery voltage of up to 12.6 V. Its maximum output current is much higher than the relay's 40 mA requirement, making it suitable for the 9 V output.

Buck Converter for 5 V

Selected component: LM2596-5.0

Input voltage range: 4.5 V – 40 V

Maximum output current: 3 A

The selected buck converter is suitable because its input voltage range covers the 12.6 V battery voltage. Its output current capability is much higher than the HC-SR04's 15 mA requirement, while its high efficiency makes it suitable for stepping down to 5 V.

//Regarding the simulation photo i will upload it in the github branch directly.

Task2:

1- Capacitors → C2

Function: Stores and releases electrical energy; filters out voltage noise and ripples.

Why is it needed : To stabilize the supply voltage and reduce noise and voltage changes.

2- Resistor → R11

Function: Limits current flow and divides voltage within the circuit.

Why is it needed : To control current and set appropriate voltage levels

3- Inductors → L1

Function: Stores energy in a magnetic field and opposes sudden changes in current.

Why is it needed : To smooth current and support efficient energy transfer, especially in switching regulator circuits.

4- Voltage Regulators → U1

Function: Maintains a stable output voltage despite changes in input voltage or load

Why is it needed : To provide a stable voltage required by the circuit's components

5- Connectors → J3

Function: Provides a physical and electrical connection between the PCB and external devices.

Why is it needed : To allow external power, signals, or data to be connected to the PCB

6- Diodes → D6

Function: Allows current to flow in only one direction while blocking reverse flow.

Why is it needed : To protect the circuit from reverse polarity or unwanted reverse current